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14 December 1987
SUPERSEDING
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(See 6.7)

MILITARY SPECIFICATION

FLUID FLOWMETER, VOLUME VELOCITY TYPE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flowmeters for use in liquids such as fresh water, sea water, fuels and oils. The flow sensor, of the turbine, vortex, or drag type, shall be located in the liquid and actuated by the motion of that liquid.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbook. The following specifications, standards, and handbook form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.

MIL-R-28803 - Readouts, Segmented, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-438 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarine Service.

MIL-STD-454 - Standard General Requirements for Electronic Equipment.

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.

MIL-STD-470 - Maintainability Program for Systems and Equipment.

MIL-STD-471 - Maintainability Verification/Demonstration/ Evaluation.

MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.

MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.

MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems Electric Power, Alternating Current. (Metric)

MIL-STD-1553 - Aircraft Internal Time Division Command/Response Multiplex Data Bus.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

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2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0518-LP-394-6000 - Navy Manufacturing Screening Program - Decrease Corporate Costs and Increase Fleet Readiness.

GENERAL SERVICES ADMINISTRATION
GSA Supply Catalog.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

(Copies of specifications, standards, handbooks, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Description. The flowmeters covered by this specification shall each consist of:

(a) A flow sensor (see 3.4) which shall be liquid velocity-responsive. Three types shall be covered:

(1) Turbine types, which produce pulses by magnetic induction as the turbine blades pass a sensing coil.
(2) Vortex types, which produce pulses as liquid vortices are shed from a specially shaped post in the liquid stream actuating a strain detector.
(3) Target types, which produce a drag or force on a target area located within the liquid flow. The force exerted on the target shall be proportional to the square of the flow rate. A strain gauge connected to the target shall produce an output analog electrical signal.

(b) An electronics unit (see 3.5) which may be combined with one of the indicator/control units (see (c) below).

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(c) Indicator/control units which shall provide the necessary indicators and controls as specified in 3.6.

3.3 General requirements. The flowmeter equipment shall be in accordance with MIL-E-16400 to the extent specified herein.

3.3.1 Parts, materials, and processes. The selection, usage, and methods of implementation of parts, materials, and processes shall be in accordance with MIL-E-16400.

3.3.2 Nonmetallic material. Nonmetals, when used for seals, protective finishes, and so forth, shall be moisture and flame resistant, shall not support fungus growth, and shall not be adversely affected by the ambient environments specified.

3.3.3 Accessibility. Parts and subassemblies which may require servicing, repair, or replacement during the life of the flowmeter equipment shall be readily accessible for repair or replacement.

3.3.4 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.3.5 Safety. Safety of flowmeter equipment shall be in accordance with requirement 1 of MIL-STD-454.

3.3.6 Painting and color. The flowmeter equipment and associated subunits shall be painted in accordance with MIL-E-16400 except for any surfaces or other parts where painting would impair meeting the requirements of this specification. Color shall be in accordance with MIL-E-16400.

3.3.7 Duty cycle. The flowmeter equipment shall be constructed for continuous operation.

3.3.8 Power input. The flowmeter equipment shall operate with type I, 115 volts, 60 hertz (Hz) electrical ungrounded power as specified in DOD-STD-1399, section 300. The equipment shall not be damaged nor require recalibration when power is interrupted and restored.

3.3.9 Electrostatic discharge caution. Where devices such as complementary metal oxide semiconductors that are sensitive to electrostatic discharge are used, a note on the printed circuit board or subassembly shall indicate to handle with special care due to the possibility of electrostatic discharge damage.

3.3.10 Low flow rate exclusion. Unless otherwise specified (see 6.2.1), flow rates below 3 ± 1 percent of rated full flow for turbine and vortex type and below 10 ± 4 percent of rated full flow for target type flowmeters shall be ignored by the flow totalizers. This characteristic prevents "sloshing" back and forth of liquid in the monitored pipe from giving false liquid transfer information.

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3.3.11 Flow rate range. The flowmeter equipment shall provide, as a minimum, the flow accuracy specified in 3.10.1. The flow range shall be as specified (see 6.2.1) and shall be marked on the flow sensor identification plate (see 3.9).

3.3.12 Multiplex data bus. When specified (see 6.2.1), the flowmeter equipment shall interface with a time division command/response multiplex data bus system in accordance with MIL-STD-1553.

3.3.13 Protection of controls and adjustments. Controls and adjustments that may be accidentally or inadvertently changed and disrupt or render inaccurate readouts and indicators shall be protected from such an inadvertent change by placement behind a cover or by requiring two operations for actuation.

3.3.14 Connectors and interconnecting cables. The subunits of the flowmeter equipment, including flow sensor, electronics unit, and indicator/control units, shall be connected to each other, as required, by interconnecting cables utilizing connectors on both ends of each cable mating to connectors on the subunits. Cables and connectors shall be as specified in MIL-E-16400. The equipment shall operate as specified when the subunits are separated from each other by up to 500 feet of cabling. Where the electronics unit is separate from the indicator/control unit, the flow sensor shall operate up to 1000 feet from the indicator/control unit. The interconnecting cable lengths shall be as specified (see 6.2.1).

3.4 Flow sensor requirements. Flow sensor requirements shall be as specified in 3.4.1 through 3.4.7.

3.4.1 Flow sensor end connections. Unless otherwise specified (see 6.2.1), flanged end connections shall be employed in accordance with MIL-STD-438 for submarines and MIL-STD-777 for surface ships.

3.4.2 Flow sensor material. The flowmeter material in contact with the liquid to be measured shall be compatible with the liquid and the flange material (see 3.4.1). MIL-STD-438 or MIL-STD-777, as applicable, and the dissimilar metal provisions of MIL-E-16400 shall determine the choice of flowmeter material.

3.4.3 Flow sensor degree of enclosure. The flow sensor unit, including the electrical connector, shall be splashproof protected as specified in MIL-STD-108 (see 4.8.7).

3.4.4 Flow sensor pressure rating. Unless otherwise specified (see 6.2.1), the flowmeter shall have a safe working pressure of 600 pounds per square inch (lb/in^2). The working pressure shall be marked on the flow sensor identification plate (see 3.9). The flowmeters shall meet the accuracy requirements of this specification at any liquid pressure from atmospheric to the flowmeter's rated operating pressure.

3.4.5 Liquid to be measured. The liquid whose flow is to be measured shall be specified (see 6.2.1) and shall be marked on the flow sensor identification plate (see 3.9).

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3.4.5.1 Liquid viscosity. The viscosity range of the liquid whose flow is to be measured shall be as specified (see 6.2.1).

3.4.5.2 Liquid operating temperature. The operating temperature range of the liquid whose flow is to be measured shall be as specified (see 6.2.1). If unspecified, the accuracy requirement shall be met over a liquid temperature range of 28 to 160 degrees Fahrenheit (°F) (see 4.8.10.1).

3.4.5.3 Specific gravity. The specific gravity of the liquid whose flow is to be measured shall be as specified (see 6.2.1).

3.4.6 Straight pipe requirement. Flow sensors shall minimize lengths of straight inlet and outlet piping required for the equipment to operate within the accuracy limits specified in 3.10.1. Required lengths of straight, unobstructed piping shall not exceed 10 pipe diameters at the inlet to the flow sensor and 5 pipe diameters at the outlet of the flow sensor. If insufficient mounting space exists for 10 diameters upstream and 5 diameters downstream, a reduced straight pipe requirement shall be as specified (see 6.2.1).

3.4.7 Pressure drop caused by the sensor. The allowable pressure drop across the sensor shall be as specified (see 6.2.1), and shall be related to flow rates.

3.5 Electronics unit. The electronics unit shall accept a signal from the flow sensor, provide signal conditioning as required, and provide outputs to the indicator/control units. The electronic unit may be a part of or separate from the indicator/control unit.

3.5.1 Maximum dimensions of electronics unit. The dimensions of the electronics unit shall be kept to a minimum consistent with required accessibility of internal parts (see 3.3.3).

3.5.2 Electronics unit degree of enclosure. The electronics unit shall be splashproof protected as specified in MIL-STD-108 (see 4.8.7).

3.5.3 Mounting of electronics unit. If the indicator/control unit is combined with the electronics unit, the resultant unit shall be panel mounted in accordance with MIL-E-16400 to provide operator access to the readouts and operating controls. If the electronics unit contains no operating controls or indicators, it shall be constructed for bulkhead mounting in accordance with MIL-E-16400.

3.6 Indicator/control units. At least one indicator/control unit shall be provided with the flowmeter equipment. If more than one indicator/control unit is required, the total number of units shall be as specified (see 6.2.1). Each different indicator/control unit shall be uniquely identified on the identification plates. Each indicator/control unit shall provide one or more of the following readouts as specified (see 6.2.1): total flow display (see 3.6.8), flow rate display (see 3.6.9), and preset batch display (see 3.6.10). The type of display (digital or analog) shall be as specified (see 6.2.1).

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3.6.1 Maximum dimension of the indicator/control units. The dimensions of the indicator/control units shall be kept to a minimum and required accessibility of internal parts (see 3.3.3).

3.6.2 Indicator/control units degree of enclosure. The indicator/control units shall be splashproof protected as specified in MIL-STD-108 (see 4.8.7).

3.6.3 Mounting of indicator/control units. The indicator/control units shall be panel mounted in accordance with MIL-E-16400 (see 3.5.3).

3.6.4 Digital displays. Digital displays shall consist of light emitting diodes (LED's) conforming to MIL-R-28803.

3.6.5 Analog displays. Analog displays shall meet the accuracy requirements of this specification unless used as duplicate indicators (for example, trend indicators).

3.6.6 Scale factor adjustment and units identification. A scale factor multiplier adjustment shall be provided with a minimum range of 0.1000 to 0.9999 with increments of 0.0001 maximum. The effect of changes in scale factor shall be linear and shall not require system calibration or cause a sensitivity change greater than 0.05 percent. If more than one display is used on one indicator/control unit, the scale factor adjustment shall cause each display to read out in the same units within the accuracy requirement of this specification. The units of the displays shall be identified by electronic data processing methods or by label plates in accordance with MIL-E-16400. Changing of the label plates (if used) shall be accomplished from the front without removal of the unit from its mounting. The scale factor adjustment shall be protected against accidental adjustment as specified in 3.3.13.

3.6.7 Self-test capability and control. A self-test capability shall be incorporated to facilitate testing of as much as possible of the electronics of the flowmeter equipment. Self-test shall be accomplished by using a fixed, accurate frequency or voltage signal. The self-test for total flow display shall not require more than 3 minutes to accomplish. There shall be means of instantaneously terminating the self-test and restoring the system to normal operation. The self-test control shall be protected against inadvertent actuation (see 3.3.13).

3.6.8 Total flow digital display. Digital display of total flow, if specified (see 6.2.1), shall be provided by the indicator/control unit. The digital display shall contain a minimum of four decimal digits. A manual reset control shall be provided to reset the total flow readout to zero. Means shall be provided to protect against accidental actuation of this control (see 3.3.13).

3.6.9 Flow rate digital display. When specified (see 6.2.1), a flow rate digital display shall be provided in the indicator/control unit. The units of this display shall be as specified in 3.6.6.

3.6.10 Preset batch readout. When specified (see 6.2.1), a preset batch readout shall be included in the indicator/control unit. The preset batch readout shall include the items specified in 3.6.10.1 through 3.6.10.5.

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3.6.10.1 Preset batch digital display. The preset batch readout shall be a digital display separate from the total flow readout. It shall consist of a minimum of four digits which display batch flow beginning at a preset amount and counting backwards to zero where it will remain until manually reset. The preset batch readout shall count at the same rate as the total flow readout.

3.6.10.2 Preset batch digital control. The preset batch digital control shall permit the operator to preselect a batch in any incremental size up to the highest number of the preset batch readout, such as 9999.

3.6.10.3 Preset batch reset controls. The preset batch reset controls shall return the preset batch readout to the preset batch quantity selected by the operator.

3.6.10.4 Preset batch complete indicator. An LED indicator in accordance with MIL-E-16400 shall be provided to indicate when the preset batch amount of liquid has been transferred. When the preset batch amount of liquid has been transferred, the preset batch readout shall reach zero and the preset batch indicator LED shall go on. The indicator shall go off when the preset batch reset control (see 3.6.10.3) is actuated.

3.6.10.5 Preset batch complete relay. A relay and associated control circuit shall be provided for remote indication by relay contact closure when the preset batch quantity of liquid has been transferred. The relay state shall match the state of the preset batch complete indicator (see 3.6.10.4). The relay contacts shall be closed when preset batch complete indicator is on and the relay contacts shall be open when preset batch complete indicator is off. The relay contacts shall be 2 pole, double throw, rated at a minimum of 120 volts, 0.5 ampere. Access to the contacts shall be via a terminal block and stuffing tube or a connector as specified in MIL-E-16400.

3.7 Reliability. The flowmeter equipment minimum mean time between failures (MTBF) shall be as specified (see 6.2.1). Reliability shall be tested in accordance with 4.8.17.

3.8 Maintenance. The flowmeter equipment shall facilitate disassembly, assembly, location of trouble sources, and calibration and maintenance without the aid of special tools. Special tools are defined as those tools not listed in the GSA Supply Catalog. Functional parts shall be readily identifiable, accessible, and removable for replacement. Functional parts are defined as individual component parts (such as discrete resistors, capacitors, and semiconductor devices) or replaceable functional assemblies (such as integrated or hybrid circuits and power supplies). For purposes of maintenance and calibration, no mechanical or electrical disassembly shall be required to gain access to functional parts except for opening a door or removal of a cover plate.

3.8.1 Program. When specified (see 6.2.1), the maintainability tasks specified in MIL-STD-470 shall be performed.

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3.8.2 Repair times. Various repair times may apply to the failure modes, but the total repair time for a maintenance event shall not exceed 2.0 hours and the mean time to repair (MTTR) shall not exceed 1.5 hours. Time to repair shall include preparation, fault location, fault correction, and check out as required but shall not include logistic time or administrative time.

3.8.3 Level of repair (LOR). All maintenance actions shall be accomplishable by no more than one repair technician of a grade no higher than E-4.

3.8.4 Maintainability demonstration. The contractor shall demonstrate a viable (to an E-4 grade technician level) means for troubleshooting each of 10 parts of the equipment which shall be considered to have failed. The mean time required to remove and replace six parts shall be measured by actual removal and replacement of these parts, one at a time, by an E-4 grade technician level or equivalent person who has been instructed by the contractor (previous to the removal and replacement time). In lieu of months of experience with the equipment, the E-4 grade or equivalent person may be instructed just prior to the removal or replacement effort for each part, but not during the actual removal and replacement. The mean time to remove or replace the failed part shall not exceed 0.8 of the MTTR specified in 3.8.2. The equipment shall operate properly after each removal or replacement. The time required to prove the equipment operates properly shall not be included in the remove or replace time measurements.

3.8.4.1 Optional maintainability demonstration. When specified (see 6.2.1), a maintainability demonstration in accordance with MIL-STD-471 shall be required. In this case, the minimum requirements of 3.8.4 may be waived.

3.9 Identification plates. Flowmeter unit major assemblies shall have identification plates in accordance with type C of MIL-P-15024 and MIL-P-15024/5. Identification plates shall be brass, black, and size 5 to 9 (larger sizes preferred). No abbreviated classification code shall be assigned. An example of a unit identification plate for the liquid flow sensing unit follows:

LIQUID FLOW SENSING UNIT

PART OF VOLUME VELOCITY LIQUID FLOWMETER

PER MIL _____

TURBINE TYPE - 3 INCH - 700 POUNDS PER SQUARE INCH (lb/in^2) WORKING

LIQUID _____

FLOW RANGE/UNITS _____

TECH. MAN. NAVSEA _____

NSN _____ CONTRACT _____

MFR. _____ MODEL NO. _____

SER. _____ INSP. _____

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Line 1 in the above example shall reflect the name of each unit major assembly to which the identification plate applies. The information on the other lines shall convey the characteristics (if different than that shown in the example) of the equipment subunit for which the identification plate applies. The form of the identification plate may be varied to best fit the mounting area available.

3.10 Performance requirements. Performance requirements shall be as specified in 3.10.1 through 3.10.16.

3.10.1 System flow accuracy. System flow accuracy shall be as specified in 3.10.1.1 through 3.10.1.2.2.

3.10.1.1 Flow rate accuracy. The flow rate indicator (see 3.6.9 and 4.8.2) accuracy shall be as follows:

(a) Turbine type: Within plus or minus 0.5 percent of the actual flow rate over the range from 10 to 100 percent of the maximum flow rate and within plus or minus 1.0 percent of the actual flow rate between 5 and 10 percent of the maximum flow rate.
(b) Vortex type: Within plus or minus 1.0 percent of the actual flow rate over the range from 10 to 100 percent of the maximum flow rate and within plus or minus 1.5 percent of the actual flow rate between 5 and 10 percent of the maximum flow rate.
(c) Target type: Within plus or minus 1.0 percent of the actual flow rate over the range from 33-1/3 to 100 percent of the maximum flow rate and within plus or minus 5 percent of the actual flow rate between 10 and 33-1/3 percent of the maximum flow rate.

3.10.1.2 Flow total accuracy. Flow total accuracy shall be as specified in 3.10.1.2.1 and 3.10.1.2.2.

3.10.1.2.1 Turbine and vortex types. For flowmeters employing pulses to represent unit volume of flow rate (turbine and vortex types), the flow totalized accuracy shall be within that accuracy specified for flow rate (see 3.10.1.1) plus an additional inaccuracy of plus or minus 2 pulse counts which shall be allowed providing the pulse count is at least 2000 per measurement, and plus or minus 1 pulse count for less than 2000 pulse counts per measurement.

3.10.1.2.2 Target type. For target flowmeters employing analog representation of unit volume of flow rate, the flow totalized (integrated) accuracy shall be within the accuracy specified for flow rate (see 3.10.1.1) plus an additional inaccuracy allowance of plus or minus 0.2 percent for the integration process.

3.10.2 Repeatability. The flowmeter readouts shall have a repeatability of plus or minus 0.2 percent plus or minus one least significant digit of the actual flow rate, total, and batch as applicable (see 4.8.4).

3.10.3 Warm-up time. The flowmeter shall attain a value within the accuracy specified in 3.10.1 in 1 minute or less after power is turned on and shall remain within this accuracy (see 4.8.9).

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3.10.4 Inclination. The flowmeter shall meet the accuracy requirement of 3.10.1 when the electronics or indicator/control units or both are subjected to an inclination of 60 degrees in accordance with the submarine requirement of MIL-E-16400 (see 4.8.5).

3.10.4.1 Sensor orientation. The flowmeter shall meet the accuracy requirement of this specification with the transducer installed on pipe which has any angular orientation from horizontal to vertical and any angular orientation with reference to the 360-degree circumference of the pipe.

3.10.5 Supply voltage and frequency (steady state). Maximum differences between readouts at the deviant supply voltage and frequency conditions specified in 4.8.11 and the nominal (115 volts, 60 Hz) at the same flow transferred shall not exceed 0.5 percent (see 4.8.11).

3.10.6 Supply voltage and frequency transients. Supply voltage and frequency transients shall be as specified in 3.10.6.1 through 3.10.6.3 (see 4.8.12).

3.10.6.1 Voltage. The total flow readout obtained by the voltage transient tests specified in 4.8.12.1 shall not differ from a reference test by more than 1 percent.

3.10.6.2 Frequency. The total flow readout obtained by the frequency transient tests specified in 4.8.12.2 shall not differ from a reference test by more than 0.5 percent.

3.10.6.3 Spike voltage. After the spike voltage test specified in 4.8.12.3, the flowmeter shall meet the accuracy requirement specified in 3.10.1.

3.10.7 Ambient temperature. The flowmeters shall pass the ambient temperature tests specified in 4.8.10. The performance criteria for acceptance of the equipment shall meet the accuracy requirement specified in 3.10.1 at all operating temperatures where a performance test is required by the tests specified in 4.8.10.

3.10.8 Humidity. The flowmeter shall meet the accuracy requirement specified in 3.10.1 for all operational checks that are performed during the humidity test specified in 4.8.13.

3.10.9 Salt fog. The flowmeter shall withstand the effects of the salt fog test specified in 4.8.8. After completion of the test and cleaning, no appreciable corrosion or other damage shall be evident. The performance of the flowmeter shall be unaffected. The flowmeter shall meet the accuracy requirement specified in 3.10.1 after the test.

3.10.10 Insulation resistance. The insulation resistance between electrical circuits and ground as determined in accordance with 4.8.3 shall be not less than 10 megohms.

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3.10.11 Vibration. The complete system shall withstand the test specified in 4.8.14. Operating and maintenance controls shall not change settings as a result of the vibration test. There shall be no evidence of mechanical or electrical damage or loosening of parts. After vibration, the flowmeter shall meet the accuracy requirements specified in 3.10.1 without adjustment of controls.

3.10.12 Shock. The complete flowmeter system shall conform to the grade A, class I, type A requirements specified in MIL-S-901. After the shock test specified in 4.8.15, but prior to any adjustments, the flowmeter shall show no deviation greater than plus or minus 5 percent between the total flow accuracy measurements made after the shock test compared to an accuracy test made before shock. After the shock test, the flowmeter shall be calibrated to the original accuracy requirement of 3.10.1 if necessary. There shall be no damage to any part of the flowmeter as a result of the shock test.

3.10.13 Production screening. Each flowmeter equipment shall be subjected to the temperature cycling and random vibration production screening specified in NAVSEA 0518-LP-394-6000 (see 4.8.2). There shall be no evidence of damage to the flowmeter equipment as a result of this test.

3.10.14 Electromagnetic susceptibility and emission control. Flowmeter equipment shall meet the following requirements specified in MIL-STD-461 (see 4.8.16): CE01, CE03, CS01, CS02, CS06, CS09, CS11, RE01, RE02, RS01, RS02, RS03, except as modified below:

(a) Requirement CS02 shall be met as specified except as follows:

(1) At frequencies between 2 megahertz (MHz) and 30 MHz, the full output of the generator (source impedence of 50 ohms, and an output capability of 12.25 volts root mean square (rms), or 3 watts, when connected to a 50 ohm load) shall be applied by common mode coupling to the test sample's power and interconnecting control lead which shall be unshielded or shall have shields grounded only at one end.

(2) The calibrating resistor, or 50 ohm load, shown in the CS02 test block diagram as specified in MIL-STD-462, shall not be connected during the test.

(b) Requirement RS03 shall be met as specified except that between the frequencies of 14 kilohertz (kHz) and 18 gigahertz (GHz), the electrical field strength test level shall be 10 volts per minute (V/m).

3.10.14.1 Susceptibility tests. During the susceptibility tests specified in 4.8.16, there shall be no deviation in readouts of the equipment when the electromagnetic interference (EMI) generator is on greater than 1 percent or 20 pounds, whichever is greater, compared to a similar readout when the EMI generator is off.

3.10.14.2 Emission tests. Emissions from the flowmeter equipment shall be within the requirements specified in MIL-STD-461 for the tests specified in 4.8.16.

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3.10.15 Digital readout toggling. The digital readouts shall be constructed so that toggling which renders the display unreadable between any two least significant or other digits shall not take place with any steady state flow input (see 4.8.6).

3.10.16 Leakage current. The flowmeter equipment shall meet the leakage current requirements specified in MIL-E-16400 (see 4.8.18).

3.11 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2).

3.12 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order an inspection system program plan shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-202.

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4.3.1 Equipment configuration. The worst case equipment configuration shall be used for all tests. For example, if flowmeter equipment is to work with interconnecting cable between transducer and electronics unit of lengths varying from 10 to 500 feet, and longer lengths are worst case, then 500 feet of cable shall be used.

4.4 First article inspection. The first article inspection shall consist of the inspections specified in table I. The inspections shall be performed in the sequence shown except for the reliability demonstration, which may be performed on other flowmeter equipments prior to, concurrent with, or after the other first article inspections. The manufacturer may perform any test at any time as a construction check.

4.4.1 Sample size. One flowmeter equipment of each type shall be subjected to first article inspection.

TABLE I. First article inspection.

Inspection	Requirement	Inspection
Group A quality conformance inspections	See table II	See table II
Inclination	3.10.4	4.8.5
Digital readout toggling	3.10.15	4.8.6
Enclosure	3.4.3, 3.5.2, and 3.6.2	4.8.7
Salt fog	3.10.9	4.8.8
Warm up time	3.10.3	4.8.9
Temperature	3.10.7	4.8.10
Supply voltage and frequency, steady state	3.10.5	4.8.11
Supply voltage and frequency, transients	3.10.6	4.8.12
Humidity	3.10.8	4.8.13
Insulation resistance	3.10.10	4.8.3
Leakage current	3.10.16	4.8.18
Vibration	3.10.11	4.8.14
Shock	3.10.12	4.8.15
EMI	3.10.14	4.8.16
Reliability	3.7	4.8.17

4.4.2 First article samples. First article samples that have passed the first article inspection specified in table I shall not be considered for shipboard installation regardless of the degree of refurbishment required. Unless otherwise directed by NAVSEA, the passed first article samples shall be retained by the first article test facility for future reference.

4.4.4 First article inspection report. When specified in the contract or order, a first article inspection report shall be prepared (see 6.2.2).

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table II in the sequence shown.

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TABLE II. Quality conformance inspection.

Inspection	Requirement	Inspection
Group A		
General examination and operation	3.2, 3.3, 3.4, 3.5, 3.6, 3.8, 3.9, and 3.10.1	4.6
Accuracy	3.10.1	4.8.1
Production screening	3.10.13	4.8.2
Insulation resistance	3.10.10	4.8.3
Leakage current	3.10.16	4.8.18
Accuracy	3.10.1	4.8.1
Repeatability	3.10.2	4.8.4
Group B		
Supply voltage and frequency, steady state	3.10.5	4.8.11
Temperature	3.10.7	4.8.10
Insulation resistance	3.10.10	4.8.3
Leakage current	3.10.16	4.8.18
Accuracy	3.10.1	4.8.1
Group C Those inspections, as determined by NAVSEA, chosen from the first article inspections specified in table I whose outcome may be affected by changes to the equipment shall be repeated for group C inspections.		

4.5.1 Inspection lot. An inspection lot shall consist of all flowmeter equipment of each type produced under essentially the same conditions and offered for inspection at the same time.

4.5.1.1 Group A inspection. All equipments in a lot shall be subjected to the group A inspections specified in table II. Equipment that fails may be corrected and retested except that no special selection of components shall be used. Any equipment or parts of equipment so corrected or reworked shall be submitted to the contracting activity or its representative for quality approval before being reinstalled in the equipment. Equipment so corrected shall be retested for all group A quality conformance tests.

4.5.1.2 Group B inspection. A random sample shall be selected from each lot that has passed the group A tests and has been subjected to the group B inspections specified in table II. The sampling procedures shall be in accordance with MIL-STD-105, with an acceptable quality level (AQL) of 1.0. Initial acquisitions shall be given a normal inspection level II. Continuation of inspection levels shall be in accordance with MIL-STD-105. If the number of nonconforming equipment equals the rejection number given as specified in MIL-STD-105, the lot shall be rejected. Nonconforming equipment may be corrected except that no special selection of components shall be used. Any equipment or parts of equipment so corrected or reworked shall be submitted to the contracting activity or its representative for quality approval before

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being reinstalled in the equipment. Corrected equipment shall be retested for all quality conformance tests except group C. Equipment so retested may be delivered if the lot is accepted and the units have suffered no damage due to handling, storage, testing or any other source.

4.5.1.3 Group C inspections. Group C inspections specified in table II shall be conducted when the construction of the equipment or the material of a vital part has been changed. One complete flowmeter equipment shall be randomly selected and subjected to the group C inspections specified in table II. Group C inspections shall be made on a sample unit which has passed the group A and B inspections. The sample unit which has been subjected to group C inspections shall not be delivered. If the sample unit fails any of the group C inspections, the sample shall be rejected. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to NAVSEA, has been taken. After corrective action has been taken, tests selected by NAVSEA from groups A, B, and C, and those tests of group C which the test sample has failed, shall be re-performed.

4.5.1.4 Quality conformance inspection report. When specified in the contract or order, a quality conformance inspection report shall be prepared (see 6.2.2).

4.6 General examination and operation. The unpowered flowmeter equipment shall be given a thorough examination to determine conformance to the requirements of this specification with respect to material, color, finish, workmanship, safety, construction, assembly, dimensions, weight, and marking of identification plates. Examination shall be limited to the examinations that may be performed without disassembling the unit in such a manner that its performance, durability, or appearance would be affected. Examination shall also include an operation check of all features and capabilities (such as self-test), operating controls, and maintenance adjustments. The operation check shall be performed at room temperature. The accuracy requirement of 3.10.1 shall also be met.

4.7 Test conditions. Except where the following factors are the variables, the tests specified in 4.8 shall be conducted with the equipment operating under the following conditions:

(a) Ambient temperature shall be 75 ± 9°F.
(b) Relative humidity shall be 60 ± 15 percent.
(c) Supply voltage shall be nominal.
(d) Supply frequency shall be nominal.
(e) Controls shall be in the neutral or normal position.

4.7.1 Test-apparatus accuracy. The accuracy of instruments and test equipment used to control and monitor the test parameters shall be in accordance with MIL-E-16400.

4.8 Tests. The flowmeter equipment and all associated test equipment shall be energized for a long enough period of time to ensure complete warm up except during the warm-up test. The flowmeter equipment shall be energized for all tests except where specified not to be energized.

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4.8.1 Accuracy. The accuracy of the flowmeter shall be checked at each of the points (percent of maximum flow rate) depending on the type of flow sensor as follows:

Turbine: 0, 5 ± 1 percent, 10 ± 1 percent, 20 ± 5 percent, 40 ± 5 percent, 60 ± 5 percent, 80 ± 5 percent, and 100 (plus 0, minus 5) percent.
Vortex: 0, 5 ± 1 percent, 10 ± 1 percent, 33-1/3 ± 5 percent, 66-2/3 ± 5 percent, and 100 (plus 0, minus 5) percent.
Target: 0, 10 ± 1 percent, 33-1/3 ± 1 percent, 50 ± 5 percent, 75 ± 5 percent, 100 (plus 0, minus 5) percent.

The period of integration for total flow shall be a minimum of 5 minutes and a maximum of 10 minutes. Accuracy shall be as specified in 3.10.1.

4.8.2 Production screening. The flowmeter equipment shall be subjected to the following temperature and vibration tests in accordance with NAVSEA 0518-LP-394-6000. The accuracy test (see 4.8.1) shall be conducted immediately following these tests and the results shall conform to 3.10.1 and 3.10.13.

4.8.2.1 Temperature cycling. With the unpowered flowmeter equipment installed in the temperature chamber, the equipment shall be subjected to three temperature cycles: Each cycle is from 77 ± 4 to minus 65 ± 4 to 158 ± 4 and back to 77 ± 4°F. The rate of temperature change shall be 20 ± 10°F per minute. A soak time of at least 30 minutes at each temperature extreme of each cycle shall be provided. After the specified number of cycles have been completed and the equipment is returned to room temperature for at least 10 minutes, the unit shall be powered. The unit shall meet the accuracy requirement of 3.10.1 and pass any self-test.

4.8.2.2 Random vibration. The flowmeter equipment shall not be energized during the random vibration. The equipment shall be mounted on the test fixture in such a manner that the axis of vibration is perpendicular to the plane of the printed circuit boards. The duration of the vibration shall be 10 minutes. The nature of the vibration and the spectrum shall be in accordance with NAVSEA 0518-LP-394-6000.

4.8.3 Insulation resistance. The insulation resistance of the unpowered flowmeter equipment shall be determined by applying 50 volts direct current (dc) between electrical inputs and cabinet ground and between electrical outputs and cabinet ground and measuring current flow. However, this shall not be done for low level inputs/outputs that may be damaged by 50 volts. Insulation resistance shall meet the requirements specified in 3.10.10.

4.8.4 Repeatability: The accuracy test of 4.8.1 shall be performed at least three times with the actual total flow measurement points controlled or determined to an accuracy at least three times better than the allowed error in 3.10.2. Repeatability shall meet the requirements specified in 3.10.2.

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4.8.5 Inclination. The flowmeter and its subunits shall be tested for accuracy in each of the following inclinations:

(a) 60 degrees from vertical forward.
(b) 60 degrees from vertical backward.
(c) 60 degrees from vertical to the left.
(d) 60 degrees from vertical to the right.
(e) Normal, horizontal position.

The requirements specified in 3.10.4 shall be met.

4.8.6 Digital readout toggling. The accuracy test of 4.8.1 shall be performed, except that at each total flow specified, flow shall be stopped and then increased (and decreased if necessary) very slowly so as to spend at least 1 minute at flow totals between least significant digits on digital electronic readouts. The toggling back and forth between the adjacent two least significant digits, if any, shall meet the requirements specified in 3.10.15.

4.8.7 Enclosure. The flowmeter equipment shall be subjected to the splashproof-protected test specified in MIL-STD-108. Performance shall meet the requirements specified in 3.4.3, 3.5.2, or 3.6.2. Accumulation of water within any equipment enclosure or failure of the equipment to operate satisfactorily shall be cause for rejection.

4.8.8 Salt fog. The flowmeter equipment shall be subjected to the salt fog test specified in MIL-STD-810, procedure I. Performance shall meet the requirements specified in 3.10.9.

4.8.9 Warm-up time. The flowmeter equipment shall be held at 32 ± 2°F for at least 2 hours. A fluid flow of 2/3 ± 5 percent of range shall be maintained and the power turned on. The warm-up time requirement of 3.10.3 shall be met.

4.8.10 Temperature. The flowmeter equipment shall be subjected to the environmental temperature tests of 4.8.10.1 and 4.8.10.2. Performance shall meet the requirements specified in 3.10.7.

4.8.10.1 Liquid temperature tests. With the flow sensor mounted in a flow test stand, the complete flowmeter shall be tested for accuracy in accordance with 4.8.1 at the minimum and maximum liquid temperatures specified in 3.4.5.2 after a minimum stabilization period of 30 minutes at each fluid temperature.

4.8.10.2 Ambient environmental temperature tests. The flowmeter equipment shall be subjected to the operating and non-operating environmental temperature test sequence specified in MIL-E-16400 for temperature range 4 (sheltered controlled environment, ship or shore) equipment, except that the upper limit of the operating temperature range shall be 140°F. The accuracy test of 4.8.1 shall be performed where an operative test is specified by the test procedures.

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4.8.11 Supply voltage and frequency (steady state). The flowmeter equipment shall be stabilized for at least 30 minutes prior to any measurements at plus 32 ± 3, plus 75 ± 4, and plus 140 ± 5°F. At each of these temperatures, the accuracy measurements of 4.8.1 shall be performed at each of the following nominal, maximum, and minimum supply steady state conditions listed below. The temperature shall be varied in steps of 20 ± 3°F with 30 minutes per step until the desired ambient is reached. The results shall meet the requirements specified in 3.10.5.

Voltage	Frequency (Hz)
115	60
123	62
107	62
107	58
123	58

4.8.12 Supply voltage and frequency (transients). The voltage and frequency transient tests shall be conducted with the flowmeter under test configured in the normal flow measurement mode and with the flow velocity adjusted to 60 ± 10 percent of range. A reference total flow measurement with a 5 ± 0.5 minute integration period shall be made at nominal voltage and frequency to be used as a standard for comparison against similar measurements which include transient tests specified in 4.8.12.1 through 4.8.12.3.

4.8.12.1 Voltage transient (type I power). Voltage transient (type I power) shall be tested in accordance with 4.8.12.1.1 and 4.8.12.1.2 and shall meet the requirements of 3.10.6.1.

4.8.12.1.1 Voltage high. The reference measurement of 4.8.12 shall be repeated with an actual flow within plus or minus 0.2 percent of the actual total flow of the reference measurement except that the supply voltage shall be 4 percent above the nominal steady state voltage through the integration period and a transient voltage of plus 16 ± 3 percent of nominal shall be superimposed recovering to the initial voltage within 2 seconds. The transient shall be applied at the halfway point (2.5 ± 0.5 minutes) of the integration period.

4.8.12.1.2 Voltage low. The reference measurement of 4.8.12 shall be repeated as in 4.8.12.1.1, except that the supply voltage shall be 4 percent below nominal throughout the integration period and a transient voltage of minus 16 ± 3 percent of nominal shall be superimposed recovering to the initial voltage within 2 seconds. The transient shall be applied at the halfway point (2.5 ± 0.5 minutes) of the integration period.

4.8.12.2 Frequency transients. Frequency transients shall be tested in accordance with 4.8.12.2.1 and 4.8.12.2.2 and shall meet the requirements of 3.10.6.2.

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4.8.12.2.1 Frequency high. The reference measurement of 4.8.12 shall be repeated with an actual flow within 0.2 percent of the actual total flow of the reference measurement except that the supply frequency shall be 1.5 percent above the nominal steady state frequency (60 Hz) and a transient frequency increase of plus 4 percent of nominal shall be superimposed recovering to the initial frequency within 2 seconds. The transient shall be applied at the halfway point (2.5 ± 0.5 minutes) of the integration period.

4.8.12.2.2 Frequency low. The reference measurement of 4.8.12 shall be repeated as in 4.8.12.2.1 except that the supply frequency shall be 1.5 percent below the nominal steady state frequency (60 Hz) and a transient frequency change of minus 4 percent of nominal shall be superimposed recovering to the initial frequency within 2 seconds. The transient shall be applied at the halfway point (2.5 ± 0.5 minutes) of the integration period.

4.8.12.3 Spike voltage. The flowmeter equipment shall be subjected to the spike voltage tested in accordance with MIL-E-16400. The performance shall meet the requirements specified in 3.10.6.3.

4.8.13 Humidity. The flowmeter equipment shall be subjected to the humidity test in accordance with MIL-STD-810, method 507.2, procedure III, except that step 6 shall be changed from 10 cycles to 5 cycles. Whenever MIL-STD-810, method 507.2 calls for an operational check, the accuracy calibration as specified in 4.8.1 shall be performed. The requirements specified in 3.10.8 shall be met.

4.8.14 Vibration. The flowmeter equipment shall be tested in accordance with type I (environmental vibration) of MIL-STD-167-1. If no resonances are observed, the 2 hour endurance test shall be conducted at 50 Hz. During the vibration test, the flowmeter shall be operating (power on) and a zero flow condition (for feasibility of test) shall be maintained. Flowmeter readouts shall be monitored periodically during the test to ascertain the readouts do not change by more than one least significant digit during the test. After the test an accuracy measurement in accordance with 4.8.1 shall be performed and performance shall meet the requirements specified in 3.10.11.

4.8.15 Shock. Prior to the shock test, the accuracy test specified in 4.8.1 shall be conducted and results recorded. An accuracy test made at the end of a previous test may be used. The flowmeter shall then be subjected to a shock test in accordance with grade A, class I, type A test of MIL-S-901. For feasibility of test, a zero flow condition shall be maintained. After the shock tests are completed, the accuracy test shall be repeated with an actual total flow accuracy within plus or minus 0.4 percent of the above. No adjustment shall be made to the flowmeter. The performance of the flowmeter shall meet the requirements specified in 3.10.12.

4.8.16 Electromagnetic interference tests. When subjected to the tests required in 3.10.14 in accordance with MIL-STD-462, modified as specified in 3.10.14, the flowmeter shall meet the requirements specified in 3.10.14.

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4.8.16.1 Susceptibility testing criteria. The flowmeter shall measure actual flow of 60 ± 10 percent of range in a test pipe, and deviation in flow (rate or total as applicable) readout shall be checked when the EMI generator is turned on compared to the same measurement when the EMI generator is off. Equivalent or alternative susceptibility test procedures may be proposed by the contractor (see 4.1.3).

4.8.16.2 Emission testing. Emission tests shall be conducted with the flowmeter operating and the flow sensor installed on a test section of pipe. There shall be an actual flow through the pipe in order to produce changes in the digital readouts. Alternative emission test procedures may be proposed by the contractor.

4.8.17 Reliability of verification. Reliability shall be tested in accordance with 4.8.17.1 or 4.8.17.2 and shall meet the requirements specified in 3.7.

4.8.17.1 Reliability demonstration. MTBF shall be demonstrated in a test program of statistically valid length using test plan IV-D or XIV-D of MIL-STD-781. The lower test MTBF (0_1) shall be as specified in 3.7 and the upper test MTBF (θ_0) shall be twice the lower test MTBF for consumer's risk (β) and producer's risk ($\propto$) of 0.2. θ_1, θ_0, β, and $\propto$ are as defined in MIL-STD-781.

4.8.17.2 Reliability prediction. Total equipment MTBF shall be verified by a reliability prediction in accordance with MIL-HDBK-217 "part stress analysis prediction", EERD-1 "Electronic Equipment Reliability Data", MIL-STD-785, task 203 - "Reliability Predictions" or combinations of the above. In accordance with the "predicted MTBF" provision in MIL-STD-781, the predicted MTBF shall be at least twice the MTBF specified in 3.7.

4.8.18 Leakage current. The leakage current between each side of the power line and the equipment enclosure, when measured in accordance with the applicable provision of MIL-E-16400, shall meet the requirements specified in 3.10.16.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C in accordance with MIL-E-17555, as specified (see 6.2.1).

5.2 Packing. Packing shall be in accordance with MIL-E-17555, level A, B, or C as specified (see 6.2.1).

5.3 Marking. In addition to any special marking (see 6.2.1), interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129. The information on the exterior and the

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outermost level of interior packaging shall include the destination, sender's address, contract or order number, and the type information required for the equipment nameplate (see 3.9).

5.4 Use of polystyrene (loose-fill) material. Use of polystyrene (loose-fill) material shall be as specified in 5.4.1 and 5.4.2.

5.4.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved (see 6.2.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler, and dunnage shall be prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labeled as follows:

"CAUTION

Contents cushioned with polystyrene (loose-fill) material
not to be taken on board ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard, or transparent flexible cellular material."

5.4.2 For level A packaging and level A and B packaging or packing. Use of polystyrene (loose-fill) material shall be prohibited for level A packaging and level A and B packing applications such as cushioning, filler, and dunnage.

6. NOTES

6.1 Intended use. Flowmeters described in this specification are intended to measure the flow rates of various liquids.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and quantity of flowmeters required.
(c) Whether low flow rates are to be ignored (see 3.3.10).
(d) Flow rate range (see 3.3.11).
(e) Whether multiplex data bus is required (see 3.3.12).
(f) Length of interconnecting cables (see 3.3.14).
(g) Whether other than flanged end connections are required (see 3.4.1).
(h) Safe working pressure, if other than as specified (see 3.4.4).
(i) Liquid to be measured (see 3.4.5).
(j) Liquid viscosity (see 3.4.5.1).
(k) Liquid operating temperature (see 3.4.5.2).
(l) Specific gravity of liquid (see 3.4.5.3).
(m) Whether reduced straight pipe requirement is needed and if so, whether the contractor is required to propose alternate solutions, such as flow straightening devices (see 3.4.6).

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(n) Number of indicator/control units required (see 3.6).
(o) Type of indicator/control unit readouts (analog or digital) (see 3.6).
  (1) Total flow digital display (see 3.6.8).
  (2) Flow rate digital display (see 3.6.9).
  (3) Preset batch capability (see 3.6.10).
(p) Minimum mean time between failures (MTBF) (see 3.7).
(q) Whether maintainability tasks are required (see 3.8.1).
(r) Whether the optional maintainability demonstration is required (see 3.8.4.1).
(s) Level of preservation-packaging (see 5.1).
(t) Level of packing (see 5.2).
(u) Special markings required (see 5.3).
(v) Whether polystyrene (loose-fill) material may be used (see 5.4.1).
(w) Pressure drop allowable (related to flow rates) (see 3.4.7).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.11	Drawings, engineering and associated lists	DI-E-7031	
4.1.2	Inspection system program plan	DI-R-4803	---
4.4.4	First article inspection report	DI-T-4902	
4.5.1.4	Inspection and test report	DI-T-5329	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Digital readouts
Drag sensor
Electromagnetic interference
Electrostatic discharge
Flow rate
Flow sensor
Liquids
Turbine sensor
Vortex sensor

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6680-N212)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-24259A(SH)	FLUID FLOWMETER, VOLUME VELOCITY TYPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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